# EECE5644 Fall 2026 – Assignment 3

Please submit your solutions in a single PDF file using the Canvas/Assignments page. The PDF should includes all math, numerical and visual results. Code can be appended in the PDF or a link to your online code repository can be included as a reference. If you point to an online repository, please do not edit the contents after the deadline, because graders may interpret a last-modified timestamp past the deadline as a late submission of the assignment. Only the contents of the PDF will be graded. Please do NOT link from the pdf to external documents online where results may be presented. Any material presented outside the PDF will not be considered for grading.

This is a graded assignment and the entirety of your submission must contain only your own work. You may benefit from publicly available literature including software (not from classmates), and AI/Coding Agents, as long as these sources are properly and clearly acknowledged and cited as references in your submission. Copying math or code from each other are not allowed and will be considered as academic dishonesty. Using any external source including AI/Coding agents without declaring in the submitted PDF will also be considered academic dishonesty. While there cannot be any written material exchange between classmates, verbal discussions to help each other are acceptable. Discussing with the instructor, the teaching assistant, classmates at open office periods, and with Generative AI tools to get clarification or to eliminate doubts are acceptable as long as they are clearly declared and cited in the submitted document.

By submitting a PDF file in response to this take home assignment you are declaring that the contents of your submission, and the associated code is your own work, except as noted in your citations to resources.

## Question 1 (60%)

In this exercise, you will train many multilayer perceptrons (MLP) to approximate the class label posteriors, using maximum likelihood parameter estimation (equivalently, with minimum average cross-entropy loss) to train the MLP. Then, you will use the trained models to approximate a MAP classification rule in an attempt to achieve minimum probability of error (i.e. to minimize expected loss with 0-1 loss assignments to correct-incorrect decisions).

**Data Distribution:** For $C = 4$ classes with uniform priors, specify Gaussian class-conditional pdfs for a 3-dimensional real-valued random vector $\mathbf{x}$ (pick your own mean vectors and covariance matrices for each class). Try to adjust the parameters of the data distribution so that the MAP classifier that uses the true data pdf achieves between $10\% - 20\%$ probability of error.

**MLP Structure:** Use a 2-layer MLP (one hidden layer of perceptrons) that has $P$ perceptrons in the first (hidden) layer with smooth-ramp style activation functions (e.g., ISRU, Smooth-ReLU, ELU, etc). At the second/output layer use a softmax function to ensure all outputs are positive and add up to 1. The best number of perceptrons for your custom problem will be selected using cross-validation.

**Generate Data:** Using your specified data distribution, generate multiple datasets: Training datasets with $100, 500, 1000, 5000, 10000$ samples and a test dataset with 100000 samples. You will use the test dataset only for performance evaluation.

**Theoretically Optimal Classifier:** Using the knowledge of your true data pdf, construct the minimum-probability-of-error classification rule, apply it on the test dataset, and empirically estimate the probability of error for this theoretically optimal classifier. This provides the aspirational performance level for the MLP classfier.

**Model Order Selection:** For each of the training sets with different number of samples, perform 10-fold cross-validation, using minimum classification error probability as the objective function, to select the best number of perceptrons (that is justified by available training data).

**Model Training:** For each training set, having identified the best number of perceptrons using cross-validation, using maximum likelihood parameter estimation (minimum cross-entropy loss) train an MLP using each training set with as many perceptrons as you have identified as optimal for that training set. These are your final trained MLP models for class posteriors (possibly each with different number of perceptrons and different weights). Make sure to mitigate the chances of getting stuck at a local optimum by randomly reinitializing each MLP training routine multiple times and getting the highest training-data log-likelihood solution you encounter.

**Performance Assessment:** Using each trained MLP as a model for class posteriors, and using the MAP decision rule (aiming to minimize the probability of error) classify the samples in the test set and for each trained MLP empirically estimate the probability of error.

**Report Process and Results:** Describe your process of developing the solution; numerically and visually report the test set empirical probability of error estimates for the theoretically optimal and multiple trained MLP classifiers. For instance show a plot of the empirically estimated test P(error) for each trained MLP versus number of training samples used in optimizing it (with semilog-x axis), as well as a horizontal line that runs across the plot indicating the empirically estimated test P(error) for the theoretically optimal classifier.

*Note*: You may use software packages for all aspects of your implementation. Make sure you use tools correctly. Explain in your report how you ensured the software tools do exactly what you need them to do.

# Question 2 (40%)

Conduct the following model order selection exercise using 10-fold cross-validation procedure and report your procedure and results in a comprehensive, convincing, and rigorous fashion:

1. Select a Gaussian Mixture Model as the true probability density function for 2-dimensional real-valued data synthesis. This GMM will have 4 components with different mean vectors, different covariance matrices, and different probability for each Gaussian to be selected as the generator for each sample. Specify the true GMM that generates data in a way that has two of the Gaussian components overlap significantly (e.g. set the distance between mean vectors comparable to the sum of their average covariance matrix eigenvalues).
2. Generate multiple data sets with independent identically distributed samples using this true GMM; these datasets will have respectively 10, 100, 1000 samples.
3. For each data set, using maximum likelihood parameter estimation principle (e.g. with the EM algorithm), within the framework of K-fold (e.g., 10-fold) cross-validation, evaluate GMMs with different model orders; specifically evaluate candidate GMMs with $1, 2, \ldots, 10$ Gaussian components. Note that both model parameter estimation and validation performance measures to be used is log-likelihood of data.
4. Repeat the experiment multiple times (e.g., at least 100 times) and report your results, indicating the rate at which each of the ten GMM orders get selected for each of the datasets you produced. Develop a good way to describe and summarize your experiment results in the form of tables/figures.